Chapter 2: Atomic Structure & Bonding

Key Equations

Average Atomic Weight: $A_{avg} = \sum f_i A_i$
(f_i: fraction of occurrence of isotope i; A_i: atomic wt of isotope i)

Force of Attraction: $F_A = \frac{|Z_1 Z_2| e^2}{4\pi\epsilon_0 r^2}$
(Z_1, Z_2: ion valences; r: interatomic distance; $e = 1.602 \times 10^{-19}$ C; $\epsilon_0 = 8.85 \times 10^{-12}$ F/m)

% Ionic Character: $\%IC = \left(1 - e^{-0.25(X_A - X_B)^2}\right) \times 100$
(X_A, X_B: electronegativities of elements A and B)

Concepts

Bohr Model: Electrons orbit nucleus in discrete energy levels ($E = hf$).
Bonding: *Ionic*—electron transfer. *Covalent*—electron sharing. *Metallic*—sea of delocalized electrons.

Chapter 3: Crystal Structures

Key Equations & Geometry

SC: $a = 2R$, 1 atom/cell. **BCC:** $a = \frac{4R}{\sqrt{3}}$, 2 atoms/cell.
FCC: $a = 2R\sqrt{2}$, 4 atoms/cell.
(a: unit cell edge length; R: atomic radius)

APF: $\frac{\text{Vol. of atoms}}{\text{Vol. of unit cell}} = \frac{n(\frac{4}{3}\pi R^3)}{a^3}$ (SC=0.52, BCC=0.68, FCC=0.74)
(n: atoms per unit cell)

Theoretical Density: $\rho = \frac{nA}{V_c N_A}$
(A: atomic weight; $V_c = a^3$: unit cell volume; $N_A = 6.022 \times 10^{23}$ atoms/mol)

Interplanar Spacing (cubic): $d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$
(h, k, l: Miller indices of the plane; a: lattice parameter)

Linear Density: $LD = \frac{\text{\# atoms centered on direction vector}}{\text{length of direction vector}}$

Planar Density: $PD = \frac{\text{\# atoms centered on plane}}{\text{area of plane}}$

Miller Indices (Planes): (1) If plane passes through origin, establish new origin. (2) Find intercepts on x, y, z. (3) Take reciprocals. (4) Clear fractions. (5) Enclose in (hkl); negatives get a bar.

Hexagonal Direction Conversion $[A\,B\,C] \to [u\,v\,t\,w]$:
$u = \frac{1}{3}(2A - B)$, $v = \frac{1}{3}(2B - A)$, $t = -(u + v)$, $w = C$

Concepts

Isotropy/Anisotropy: Isotropic—properties independent of direction; anisotropic—vary with direction.

Chapter 4: Imperfections in Solids

Key Equations

Total Atomic Sites: $N = \frac{N_A \rho}{A}$
(N: atomic sites per unit volume; ρ: density; A: atomic weight)

Equilibrium Vacancies: $N_v = N \exp\left(-\frac{Q_v}{kT}\right)$
(N_v: # vacancies; Q_v: activation energy for vacancy formation; $k = 8.62 \times 10^{-5}$ eV/atom·K; T: temp in K)

Mass to Weight %: $C_1 = \frac{m_1}{m_1 + m_2} \times 100$
(m_1, m_2: masses of elements 1 and 2)

Weight % to Atom %: $C_1' = \frac{C_1/A_1}{C_1/A_1 + C_2/A_2} \times 100$
(C_1': atom % of element 1; A_1, A_2: atomic weights)

Atom % to Weight %: $C_1 = \frac{C_1' A_1}{C_1' A_1 + C_2' A_2} \times 100$

Average Density (2-component alloy): $\rho_{ave} = \frac{100}{\frac{C_A}{\rho_A} + \frac{C_B}{\rho_B}}$
(C_A, C_B: wt% of components A, B; ρ_A, ρ_B: densities of A, B)

Average Atomic Weight (2-component alloy): $A_{ave} = \frac{100}{\frac{C_A}{A_A} + \frac{C_B}{A_B}}$
(A_A, A_B: atomic weights of A, B)

Atoms/Volume of Element 1 in Alloy: $N_1 = \frac{N_A C_1}{\frac{C_1 A_1}{\rho_1} + \frac{A_1(100 - C_1)}{\rho_2}}$
(N_1: atoms of element 1 per cm^3; C_1: wt% of element 1; ρ_1, ρ_2: densities)

Concepts

Vacancy: Atom missing from lattice site. **Self-Interstitial:** Extra atom in void space.

Chapter 5: Diffusion

Key Equations

Fick's First Law (Steady-State): $J = -D\frac{\Delta C}{\Delta x}$
(J: diffusion flux [kg/m^2·s]; D: diffusion coefficient [m^2/s]; C: concentration; x: distance)

Mass Diffused: $M = JAt = -DAt\frac{\Delta C}{\Delta x}$
(M: mass diffused; A: cross-sectional area; t: time)

Fick's Second Law (Non-Steady-State):
$\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$
(C_x: conc. at depth x and time t; C_0: initial bulk conc.; C_s: surface conc.)

Diffusion Coefficient: $D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$
(D_0: pre-exponential [m^2/s]; Q_d: activation energy [J/mol]; $R = 8.31$ J/mol·K; T: temp in K)

Activation Energy (Two Temps): $Q_d = -R\frac{\ln D_1 - \ln D_2}{1/T_1 - 1/T_2}$

erf Table & Interpolation: $\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$

z	erf	z	erf	z	erf	z	erf
0.00	0.000	0.30	0.329	0.60	0.604	1.00	0.843
0.10	0.112	0.40	0.428	0.70	0.678	1.20	0.910
0.20	0.223	0.50	0.521	0.80	0.742	1.50	0.966
0.25	0.276	0.55	0.563	0.90	0.797	2.00	0.995

Chapter 6: Mechanical Properties

Key Equations

Eng. Stress: $\sigma = F/A_0$ **Eng. Strain:** $\epsilon = \Delta l / l_0$
(F: applied force; A_0: original cross-sectional area; Δl: change in length; l_0: original length)

Shear: $\tau = F/A_0$, $\gamma = \tan\theta$ (θ: shear angle)
Hooke's Law: $\sigma = E\epsilon$ (elastic only) $\tau = G\gamma$
(E: elastic/Young's modulus; G: shear modulus)

Elastic Strain from geometry: $\epsilon_z = \sigma/E = F/(A_0 E)$
Poisson's Ratio: $\nu = -\epsilon_x/\epsilon_z = -\epsilon_y/\epsilon_z$
(ϵ_x, ϵ_y: lateral strains; ϵ_z: axial strain)

True Stress/Strain: $\sigma_T = \sigma(1 + \epsilon)$; $\epsilon_T = \ln(1 + \epsilon)$
Strain Hardening: $\sigma_T = K\epsilon_T^n$
(K: strength coefficient; n: strain-hardening exponent)

Resilience: $U_r \approx \frac{1}{2}\sigma_y \epsilon_y = \frac{\sigma_y^2}{2E}$
(σ_y: yield strength; ϵ_y: strain at yield)

Elastic Elongation: $\Delta l = \frac{l_0 F}{A_0 E}$

Concepts

Yield Strength: 0.002 strain offset. **Tensile Strength:** Max on engineering curve. **% Elongation:** $\frac{l_f - l_0}{l_0} \times 100$ (l_f: final length).

Chapter 7: Dislocations & Strengthening

Key Equations

Resolved Shear Stress: $\tau_R = \sigma \cos\phi \cos\lambda$
(ϕ: angle between stress axis and slip plane normal; λ: angle between stress axis and slip direction)

Yield Strength from CRSS: $\sigma_y = \frac{\tau_{CRSS}}{(\cos\phi\cos\lambda)_{\max}}$
(τ_{CRSS}: critical resolved shear stress — minimum shear stress to initiate slip)

Hall-Petch: $\sigma_y = \sigma_0 + k_y d^{-1/2}$
(σ_0: friction stress; k_y: strengthening coefficient; d: avg grain diameter)

% Cold Work: $\%CW = \left(\frac{A_0 - A_d}{A_0}\right) \times 100$
(A_0: original area; A_d: deformed cross-sectional area)

Concepts

Slip System: Slip plane + slip direction. **Strain Hardening:** Ductile metal gets harder/stronger as plastically deformed (more dislocations, which impede each other).

Chapter 8: Failure

Key Equations

Stress Concentration: $\sigma_m = 2\sigma_0\sqrt{a/\rho_t} = K_t\sigma_0$
(σ_0: nominal applied stress; a: half internal flaw length; ρ_t: crack tip radius; K_t: stress concentration factor)

Critical Stress (Brittle): $\sigma_c = \sqrt{\frac{2E\gamma_s}{\pi a}}$
(γ_s: specific surface energy; a: half crack length)

Fracture Toughness: $K_c = Y\sigma_c\sqrt{\pi a}$
(K_c: fracture toughness [MPa$\sqrt{\text{m}}$]; Y: dimensionless geometric factor)

Cyclic Parameters: $\Delta\sigma = \sigma_{\max} - \sigma_{\min}$; $\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}$; $R = \frac{\sigma_{\min}}{\sigma_{\max}}$
($\Delta\sigma$: stress range; σ_m: mean stress; R: stress ratio)

Creep Rate: $\dot{\epsilon}_s = K_2\sigma^n \exp(-Q_c/RT)$
($\dot{\epsilon}_s$: steady-state creep rate; K_2, n: material constants; Q_c: creep activation energy)

Larson-Miller: $m = T(C + \log t_r)$
(T: temp in K or R; C: material constant; t_r: time to rupture)

Concepts

Fatigue: Structural damage under cyclic stress. **S-N Curve:** Fatigue lifetime = cycles to failure at given σ; fatigue strength = σ for given number of cycles.

Chapter 9: Phase Diagrams

Key Equations — Lever Rule

For a two-phase region with tie line from C_α to C_β at overall composition C_0:
$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha}$ $W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha}$
(W_α, W_β: mass fractions of α and β phases)

Eutectic Microconstituent: $W_e = \frac{C_0' - 18.3}{43.6}$ (Pb-Sn)
Fe-C Hypoeutectoid ($C_0' < 0.76$ wt%C):
$W_p = \frac{C_0' - 0.022}{0.74}$ $W_{\alpha'} = \frac{0.76 - C_0'}{0.74}$
(W_p: fraction pearlite; $W_{\alpha'}$: fraction proeutectoid ferrite)
Fe-C Hypereutectoid ($C_0' > 0.76$ wt%C):

$W_p = \frac{6.70 - C_1'}{5.94}$ $\quad W_{Fe_3C'} = \frac{C_1' - 0.76}{5.94}$
($W_{Fe_3C'}$: fraction proeutectoid cementite)

Fe-C Key Points

Eutectoid: $\gamma(0.76\%C) \rightarrow \alpha(0.022\%C) + Fe_3C(6.7\%C)$ at 727°C
Eutectic: $L \rightarrow \gamma + Fe_3C$ at 1147°C, 4.30 wt%C
Phases: α-ferrite (BCC, soft); austenite γ (FCC); cementite Fe_3C (hard, 6.7%C)
Hypoeutectoid (<0.76%C): proeutectoid α + pearlite
Hypereutectoid (>0.76%C): proeutectoid Fe_3C + pearlite

Reactions (Definitions)

Eutectic: $L \rightarrow \alpha + \beta$ (liquid → two solids)
Eutectoid: $\gamma \rightarrow \alpha + Fe_3C$ (solid → two solids)
Peritectic: $L + \alpha \rightarrow \beta$ (liquid + solid → solid)

Example: Fe-C Lever Rule (HW7)

0.45 wt%C alloy (hypoeutectoid) at room temp:
$W_p = \frac{0.45 - 0.022}{0.74} = 0.578 \quad W_{\alpha'} = \frac{0.76 - 0.45}{0.74} = 0.419$

Chapter 10: Phase Transformations

Key Equations

Critical Radius for Nucleation:
$r^* = -\frac{2\gamma T_m}{\Delta H_f} \cdot \frac{1}{T_m - T} = \frac{-2\gamma}{\Delta G_v}$
(γ: solid-liquid interfacial energy [J/m^2]; ΔH_f: latent heat of fusion [J/m^3]; T_m: melting temp [K]; T: nucleation temp [K]; ΔG_v: free energy change per unit volume)
Activation Free Energy:
$\Delta G^* = \frac{16\pi\gamma^3 T_m^2}{3\Delta H_f^2} \cdot \frac{1}{(T_m - T)^2}$
Avrami Equation: $y = 1 - \exp(-kt^n)$
(y: fraction transformed; k, n: material-specific constants; t: time)
Transformation Rate: rate $= 1/t_{0.5}$ ($t_{0.5}$: time for 50% transformation)

Example: Critical Radius (HW9, 10.3)

Cu: $T_m = 1085$°C, nucleates at 849°C, $\Delta H_f = -1.77 \times 10^9$ J/m^3, $\gamma = 0.200$ J/m^2:
$r^* = \frac{-2(0.200)(1085+273)}{-1.77\times10^9} \cdot \frac{1}{1085-849} = 1.30 \times 10^{-9}$ m

Example: Avrami Rate (HW9, 10.8)

$n = 2.5$, $y = 0.40$ at $t = 200$ min. Find k: $k = \frac{-\ln(1-y)}{t^n} = \frac{-\ln(0.60)}{200^{2.5}} = 9.0 \times 10^{-8}$.
Find $t_{0.5}$: $t_{0.5} = \left[\frac{-\ln(0.5)}{k}\right]^{1/n} = \left[\frac{0.693}{9.0\times10^{-8}}\right]^{1/2.5} = 311$ min; rate $= 1/311 = 3.2 \times 10^{-3}$ min^{-1}

Chapter 12: Ceramics

Key Equations

Theoretical Density: $\rho = \frac{n'(\Sigma A_C + \Sigma A_A)}{V_C N_A}$
(n': formula units/cell; A_C: sum of cation atomic weights; A_A: sum of anion atomic weights; V_C: unit cell volume)
Frenkel Defect #: $N_{fr} = N \exp\left(-\frac{Q_{fr}}{2kT}\right)$
(N_{fr}: # Frenkel defects; Q_{fr}: activation energy for Frenkel formation)
Schottky Defect #: $N_s = N \exp\left(-\frac{Q_s}{2kT}\right)$
(N_s: # Schottky defect pairs; Q_s: activation energy for Schottky formation)

Concepts

Frenkel Defect: Cation vacancy + cation interstitial pair (cation moves from lattice site to void; charge neutral).
Schottky Defect: Paired cation vacancy + anion vacancy (maintains charge neutrality).
Ceramic Structures (n'):
Rock Salt (NaCl, FeO, MgO): $n' = 4$ Cesium Chloride (CsCl): $n' = 1$
Zinc Blende (ZnS, SiC): $n' = 4$ Fluorite/Antifluorite: $n' = 4$
Perovskite: $n' = 1$ Spinel: $n' = 8$

Example: Schottky (HW9, 12.29)

NaCl at 801°C ($T = 1074$ K), $Q_s = 2.3$ eV:
$N_s/N = \exp\left(\frac{-2.3}{2(8.62\times10^{-5})(1074)}\right) = \exp(-12.43) = 3.97 \times 10^{-6}$

Chapter 14: Polymers

Key Equations

Number-Avg Mol. Weight: $M_n = \sum x_i M_i$
(x_i: number fraction of chains in size range i; M_i: mean mol. wt of range i)
Weight-Avg Mol. Weight: $M_w = \sum w_i M_i$
(w_i: weight fraction of chains in size range i)
Degree of Polymerization: $DP = M_n/m$
(m: molecular weight of repeat unit [g/mol])
Repeat Unit MW (copolymer): $m = \sum f_j m_j$
(f_j: mole fraction of monomer j; m_j: mol. wt of repeat unit j)

Example (HW10)

Polypropylene $M_n = 10^6$ g/mol, repeat unit C_3H_6:
$m = 3(12.01) + 6(1.01) = 42.09$ g/mol; $DP = 10^6/42.09 = 23,760$

Chapter 18: Electrical Properties

Key Equations

Ohm's Law: $V = IR$ **Resistivity:** $\rho = RA/l$ **Conductivity:** $\sigma = 1/\rho$
(V: voltage [V]; I: current [A]; R: resistance [Ω]; A: cross-sectional area; l: length)
Current Density: $J = \sigma\mathcal{E}$ **E-field:** $\mathcal{E} = V/l$
Metal Conductivity: $\sigma = n|e|\mu_e$
(n: electron concentration [m^{-3}]; $|e| = 1.6 \times 10^{-19}$ C; μ_e: electron mobility [m^2/V·s])
Matthiessen's Rule: $\rho_{total} = \rho_t + \rho_i + \rho_d$
(ρ_t: thermal contribution; ρ_i: impurity contribution; ρ_d: deformation contribution)
Intrinsic Semiconductor: $n = p = n_i$; $\sigma = n_i|e|(\mu_e + \mu_h)$
(n_i: intrinsic carrier concentration; p: hole concentration; μ_h: hole mobility)
n-type extrinsic: $\sigma \cong n|e|\mu_e$ **p-type:** $\sigma \cong p|e|\mu_h$
Capacitance (vacuum): $C = \epsilon_0 A/l$ $\epsilon_0 = 8.85 \times 10^{-12}$ F/m
Capacitance (dielectric): $C = \epsilon A/l = \epsilon_r\epsilon_0 A/l$
(ϵ_r: relative permittivity/dielectric constant; $\epsilon = \epsilon_r\epsilon_0$: permittivity)
Charge: $Q = CV$ (Q: charge [C]; C: capacitance [F])

Constants

Si: $\mu_e = 0.14$ m^2/V·s, $\mu_h = 0.048$ m^2/V·s, $n_i = 1.1 \times 10^{16}$ m^{-3}
Ge: $\mu_e = 0.38$, $\mu_h = 0.18$ m^2/V·s, $n_i = 2.4 \times 10^{19}$ m^{-3}

Concepts

Intrinsic Semiconductor: Conductivity from pure material; $n = p$ (e.g., pure Si, Ge).
Extrinsic Semiconductor: Conductivity controlled by dopants. *n-type* = donor impurity (Group V, e.g., P in Si), excess electrons. *p-type* = acceptor impurity (Group III, e.g., B in Si), excess holes.

Example: Intrinsic Carrier Concentration (HW10)

GaAs: $\sigma = 3 \times 10^{-7}$ (Ω·m)$^{-1}$, $\mu_e = 0.80$, $\mu_h = 0.04$ m^2/V·s:
$n_i = \frac{\sigma}{|e|(\mu_e + \mu_h)} = \frac{3\times10^{-7}}{(1.6\times10^{-19})(0.84)} = 2.2 \times 10^{12}$ m^{-3}

Example: Extrinsic Conductivity (HW10)

Ge doped with 5×10^{22} m^{-3} Sb (n-type), $\mu_e = 0.1$, $\mu_h = 0.05$ m^2/V·s:
$\sigma \cong n|e|\mu_e = (5 \times 10^{22})(1.6 \times 10^{-19})(0.1) = 800$ (Ω·m)$^{-1}$

Example: Capacitance

Parallel plate, $A = 2 \times 10^{-4}$ m^2, $l = 1 \times 10^{-3}$ m, $\epsilon_r = 6.0$:
$C = (6.0)(8.85 \times 10^{-12})(2 \times 10^{-4})/(1 \times 10^{-3}) = 1.06 \times 10^{-12}$ F

Chapter 19: Thermal Properties

Key Equations

Heat Capacity: $C = dQ/dT$ (J/mol·K)
(C: molar heat capacity; Q: heat added; T: temperature)
Specific Heat: $c = dQ/(m\,dT)$ (J/kg·K)
(c: specific heat per unit mass; m: mass)
Linear Thermal Expansion: $\Delta l/l_0 = \alpha_l \Delta T$
(α_l: linear thermal expansion coefficient [°C^{-1} or K^{-1}]; $\Delta T = T_f - T_0$)
Volume Expansion: $\Delta V/V_0 = \alpha_v \Delta T \approx 3\alpha_l \Delta T$
Heat Flux (Fourier's Law): $q = -k\,dT/dx$
(q: heat flux [W/m^2]; k: thermal conductivity [W/m·K]; dT/dx: temperature gradient)
Thermal Stress: $\sigma = E\alpha_l(T_0 - T_f)$
(σ: thermal stress; E: elastic modulus; T_0: initial temp; T_f: final temp)

Concepts

Heat Capacity (C): Energy to raise 1 mol by 1 K.
Specific Heat (c): Same but per unit mass. $C_p > C_v$ always.
Thermal Conductivity: Metals highest (free e^-), ceramics moderate (phonons), polymers lowest.
Thermal Shock Resistance: $TSR \propto \sigma_f k/(E\alpha_l)$; high TSR = resists fracture from rapid ΔT.

HW Problem Reference

Phase Diagram — Compositions at a Temperature (HW7)

Locate tie line at temperature. Read C_α, C_β from phase boundaries; apply lever rule for W_α, W_β.

Resistivity Calculation (HW10)

$\rho = RA/l$. Wire: $A = \pi r^2$.
Ex: $A = \pi(0.001)^2 = 3.14 \times 10^{-6}$ m^2, $R = 0.003$ Ω, $l = 0.3$ m: $\rho = 3.14 \times 10^{-8}$ Ω·m.

Ceramic Density (HW9, 12.16)

NaCl ($n' = 4$, $A_{Na} = 22.99$, $A_{Cl} = 35.45$, $a = 0.562$ nm):
$\rho = \frac{4(22.99+35.45)}{(0.562\times10^{-9})^3(6.022\times10^{23})} = 2.16 \times 10^3$ kg/m^3

Polymer MW (HW10)

$M_n = \sum x_i M_i$; $M_w = \sum w_i M_i$; $DP = M_n/m$. Use midpoint of each range as M_i.

Nucleation (HW9)

$k = \frac{-\ln(1-y)}{t^n}$; then $t_{0.5} = \left[\frac{\ln 2}{k}\right]^{1/n}$; rate $= 1/t_{0.5}$.

Common Constants

$R = 8.314$ J/mol·K, $k_B = 8.62 \times 10^{-5}$ eV/atom·K, $N_A = 6.022 \times 10^{23}$ mol^{-1}, $|e| = 1.6 \times 10^{-19}$ C, $\epsilon_0 = 8.85 \times 10^{-12}$ F/m